

How to Prepare a Good Poster?

JUNE 2, 2026

“The more strikingly visual your presentation is, the more people will remember it. And more importantly, they will remember you.”

– Paul Arden



What is a Poster?

A poster is a visual and textual presentation that summarizes a research project, idea, or topic on a single large sheet.

It combines:

- Text,
- Graphics/Pictures
- Structure to communicate the essence of your work quickly and clearly.



Poster vs Talk

Poster	Talk
spatial, self-paced, and <i>interactive</i>	sequential, time-based explanation

Structure and Design

Partition poster into well-defined sections.

- Content determines structure.
- Flow from top to bottom, left to right.
- Common structure: Intro -> Background -> Results -> Discussion/Future Research
- *Abstract is not necessary for poster*



Recommended Poster Sections

- **Title:** Clear and Descriptive
 - **Introduction/Motivation:** Why does this topic matter? What question(s) are you asking?
 - **Data:** Where did the data come from? How many observations/variables?
 - **Methods:** What analysis did you run in R? (e.g. regression, visualization, clustering)
 - **Results:** Your key findings
 - **References:** Credit your data source and any tools you used
 - **Acknowledgements:** Credit your mentors and anyone else you would like to thank
- 

Structure and Design

From up close or afar, the poster should be inviting.

- Avoid excessive text.
- Aim for approximately 20% text, 40% visuals, and 40% empty space.
- Nothing wrong with a little white space. Break up text with figures, pictures and equations.
- Use short blocks of text or bullet points.



Structure and Design

Design for readability

- Choose sans serif fonts like Arial or Verdana. (No more than two types).
- Avoid switching fonts to highlight key points (**bold** or *italics*).
- Use high contrast between text and background (avoid red/green combinations).
- Use no more than 2-3 colors.
- Make sure titles and headings stand out by using larger font sizes than the rest of the text

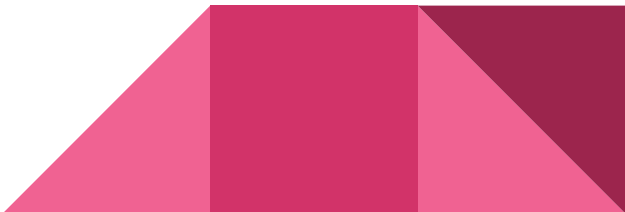
Title: 85 pt

Authors: 56 pt

Sub-headings: 36 pt

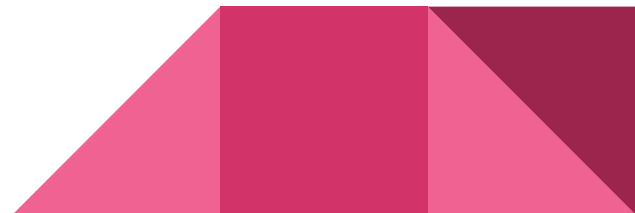
Body text: 24 pt

Captions: 18 pt



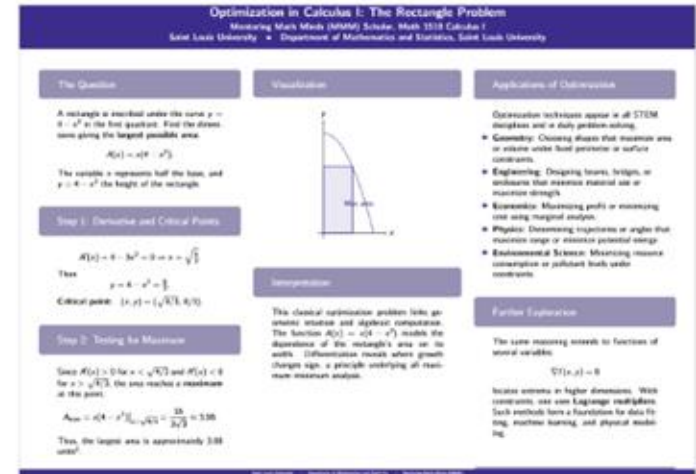
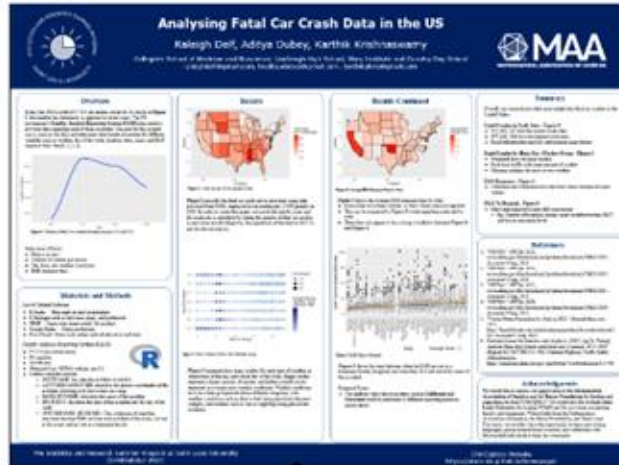
Using R Visuals in Your Poster

- Export directly from R, do not take screenshots (blurry when printed)
- Export at high resolution
- Make axis labels large enough to read from 3 feet away
- Always label axes with clear names and units
- Avoid red/green combinations



First Impressions

Which of the two presentations would you rather attend? Why?



Logistics

- **Dimensions:** 48" wide x 36" tall (landscape orientation)
- **Template:** A PowerPoint poster template will be provided, do not change the dimensions
- **Preparation Due Date:** Make sure to prepare your posters by **Thursday, June 11th, 3PM**
- **Presentation date:** **Friday, June 12th, 1-2 PM**
- **Display:** You will stand next to your poster and explain your work to audience



Discussion Time

For each of the following posters, ask yourself the following questions:

- How does the **structure** help communicate the work?
 - Are the sections well-defined?
 - Does the poster feel cluttered?
 - Are you interested in reading more after the initial glance?
- Is the **content** effective?
 - What is the main subject of the poster?
 - Who is the intended audience for this poster?
 - Can the poster be easily summarized?

Optimization in Calculus I: The Rectangle Problem

Mentoring Math Minds (MMM) Scholar, Math 1510 Calculus I
Saint Louis University • Department of Mathematics and Statistics, Saint Louis University

The Question

A rectangle is inscribed under the curve $y = 4 - x^2$ in the first quadrant. Find the dimensions giving the **largest possible area**.

$$A(x) = x(4 - x^2).$$

The variable x represents half the base, and $y = 4 - x^2$ the height of the rectangle.

Step 1: Derivative and Critical Points

$$A'(x) = 4 - 3x^2 = 0 \Rightarrow x = \sqrt{\frac{4}{3}}.$$

Then

$$y = 4 - x^2 = \frac{8}{3}.$$

Critical point: $(x, y) = (\sqrt{4/3}, 8/3)$.

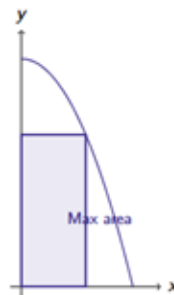
Step 2: Testing for Maximum

Since $A'(x) > 0$ for $x < \sqrt{4/3}$ and $A'(x) < 0$ for $x > \sqrt{4/3}$, the area reaches a **maximum** at this point.

$$A_{\max} = x(4 - x^2)|_{x=\sqrt{4/3}} = \frac{16}{3\sqrt{3}} \approx 3.08.$$

Thus, the largest area is approximately 3.08 units².

Visualization



Interpretation

This classical optimization problem links geometric intuition and algebraic computation. The function $A(x) = x(4 - x^2)$ models the dependence of the rectangle's area on its width. Differentiation reveals where growth changes sign, a principle underlying all maximum–minimum analysis.

Applications of Optimization

Optimization techniques appear in all STEM disciplines and in daily problem-solving.

- **Geometry:** Choosing shapes that maximize area or volume under fixed perimeter or surface constraints.
- **Engineering:** Designing beams, bridges, or enclosures that minimize material use or maximize strength.
- **Economics:** Maximizing profit or minimizing cost using marginal analysis.
- **Physics:** Determining trajectories or angles that maximize range or minimize potential energy.
- **Environmental Science:** Minimizing resource consumption or pollutant levels under constraints.

Further Exploration

The same reasoning extends to functions of several variables:

$$\nabla f(x, y) = 0$$

locates extrema in higher dimensions. With constraints, one uses **Lagrange multipliers**. Such methods form a foundation for data fitting, machine learning, and physical modeling.

Collection and Analysis of Weather Forecast Error Data

Project Goals

- 1. Acquire U.S. weather forecast and city feature data.
- 2. Organize the acquired data so that it contains weather observations alongside predicted values at regular time intervals.
- 3. Use this data to create visualizations and models that seek to explain which areas of the U.S. struggle with temperature prediction and the possible explanations for these forecasting difficulties.

Data Acquisition and Organization

The process for weather data collection is as follows:

- 1. Receive emails from `wx-natal@lists.illinois.edu` that include weather data (forecasts and observations) for January 30, 2021 through February 1, 2022.
- 2. Organize the email files into an R data frame that has a similar format to that of the emails but that excludes irrelevant information.
- 3. Create a reorganized data frame in which each row includes all of the relevant observations and forecasts for a unique (date, city) pair. Also create an expanded data frame that has eight rows for each unique (date, city) pair.
- 4. Add a possible error column to the expanded email data.

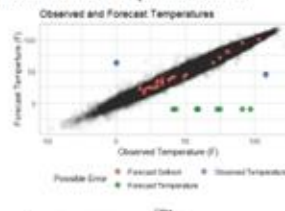
We also gathered the following information on each city for which we have weather data:

- 1. longitude and latitude coordinates,
- 2. Koppen classification,
- 3. elevation,
- 4. distance to coastline,
- 5. mean wind speed, and
- 6. average annual precipitation.

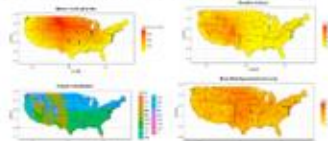
These features help us determine why some areas of the country have larger temperature forecast error than others.

Data Visualization

Below is a scatter plot of observed temperatures vs. forecast temperatures. The colored points correspond to observations that are possible errors, and the color indicates the problem variable.



Above is a map of the cities for which we have gathered weather data and below are some maps showing the different features we collected for each city.



Statistical Tests

We perform paired t-tests to determine whether or not predictions improve as the number of hours before observation decreases. In this analysis, we use a 5% significance level and only consider the cities in the continental United States. The results for the tests examining high (top table) and low (bottom table) forecast errors follow.

Three numbers are reported for each test: the number of cities in which predictions (1) improved, (2) got worse, and (3) neither improved nor got worse.

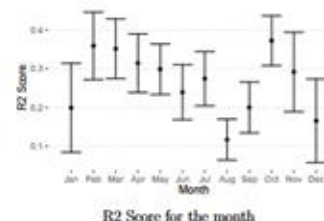
	12 hrs	24 hrs	36 hrs
48 hrs	159 0 0	146 0 13	110 0 49
36 hrs	158 0 1	110 0 49	NA
24 hrs	139 0 20	NA	NA

	12 hrs	24 hrs	36 hrs
48 hrs	141 0 18	134 0 25	56 0 103
36 hrs	121 0 38	85 0 74	NA
24 hrs	57 1 101	NA	NA

Modeling

To determine which predictors are the most significant in predicting the forecast error, we use stepAIC - a variable selection method that gives the best model with the fewest predictors that still has good predictive capabilities. We run this experiment at the season level and month level. We perform an 80/20 train/test split repeated 50 times. In the table below, we summarize the percentage of time each predictor was chosen.

Predictor	% of times chosen
Distance to coast	77.0
Latitude	70.0
Koppen classification	56.3
Elevation	37.94
Wind	31.67



To predict the error in a specific city on a given day, we predict the error on the daily rain, season, and outlook. We observe that this model produced sub-optimal results.

Kriging

Kriging is a method of interpolation used in geo-statistics. The method uses the preexisting data to create a variogram model of temperature error. A variogram (γ) is a function that describes the spatial dependence of a variable in a space. There are several different models used, but this project utilized the exponential model given by the equation:

$$\gamma(h) = b + C_0 * (1 - e^{-a * h}) \quad \text{where} \quad \begin{array}{l} b \text{ Nugget} \\ C_0 \text{ Sill} \\ a \text{ Range} \\ h \text{ Distance} \end{array}$$

Through this, the mean temperature error of any location in the U.S. without recorded data can be predicted by its neighbors based on their spatial proximity.



Project Conclusions

- 1. A larger difference between the prediction time and observation time generally corresponds to more cities in which temperature predictions improve.
- 2. Latitude and distance to coast are important predictors.
- 3. The data we have produced and managed is a valuable contribution for further study to be conducted.



Southern Flounder Exhibit Temperature-Dependent Sex Determination



Introduction

Southern flounder (*Paralichthys lethostigma*) support valuable fisheries and show great promise for aquaculture. Female flounder are known to grow faster and reach larger adult sizes than males. Therefore, information on sex determination that might increase the ratio of female flounder is important for aquaculture.

Objective

This study was conducted to determine whether southern flounder exhibit temperature-dependent sex determination (TSD), and if growth is affected by rearing temperature.

Methods

- Southern flounder broodstock were strip spawned to collect eggs and sperm for *in vitro* fertilization.
- Hatched larvae were weaned from a natural diet (rotifers/*Artemia*) to high protein pelleted feed and fed until satiation at least twice daily.
- Upon reaching a mean total length of 40 mm, the juvenile flounder were stocked at equal densities into one of three temperatures 18, 23, or 28°C for 245 days.
- Gonads were preserved and later sectioned at 2-6 microns.
- Sex-distinguishing markers were used to distinguish males (spermatogenesis) from females (oogenesis).

Histological Analysis

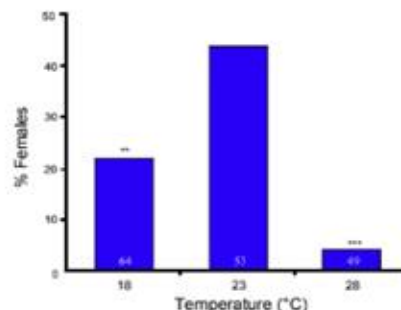


Male Differentiation

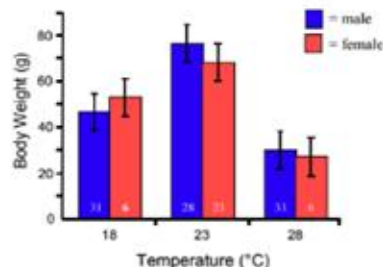


Female Differentiation

Temperature Affects Sex Determination



Growth Does Not Differ by Sex



Results

- Sex was discernible in most fish greater than 120 mm long.
- High (28°C) temperature produced 4% females.
- Low (18°C) temperature produced 22% females.
- Mid-range (23°C) temperature produced 44% females.
- Fish raised at high or low temperatures showed reduced growth compared to those at the mid-range temperature.
- Up to 245 days, no differences in growth existed between sexes.

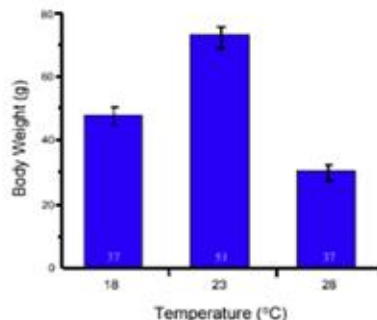
Conclusions

- These findings indicate that sex determination in southern flounder is temperature-sensitive and temperature has a profound effect on growth.
- A mid-range rearing temperature (23°C) appears to maximize the number of females and promote better growth in young southern flounder.
- Although adult females are known to grow larger than males, no difference in growth between sexes occurred in age-0 (< 1 year) southern flounder.

Acknowledgements

The authors acknowledge the Saltonstall-Kennedy Program of the National Marine Fisheries Service and the University of North Carolina Sea Grant College Program for funding this research. Special thanks to Lea Ware and Beth Shamps for help with the work.

Rearing Temperature Affects Growth



What Makes a Good Poster?

1. Focus on One Story

Communicate the path of discovery

1. Logical Structure

*Title and Authors, Motivation & Problem, Methods & Key Ideas, Conclusion & Perspectives,
Acknowledgments, References*

1. Visual Hierarchy & Readability

2. Mathematical Content

Show the idea or insight instead of all the algebra/calculation/theory



Presenting Your Poster

Structure your 60 seconds pitch like this:

- **Motivation:** “We looked at [topic] because...”
- **Question:** “Our research question was...”
- **What you did:** “Using R, we analyzed [dataset] and...”
- **What you found:** “The most interesting finding was...”
- **Why it matters:** “This could mean that...”

Tips:

- Make eye contact! Don't just read off the poster
 - Point to your visuals as you talk through results. But do not block the poster the whole time
 - Practice out loud at least twice before the event
- 



Poster Template QR Code

Thank You!

